

# OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 2

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Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



# Preparing for Exploratory Data Analysis

**Tools:** - laptop

- git

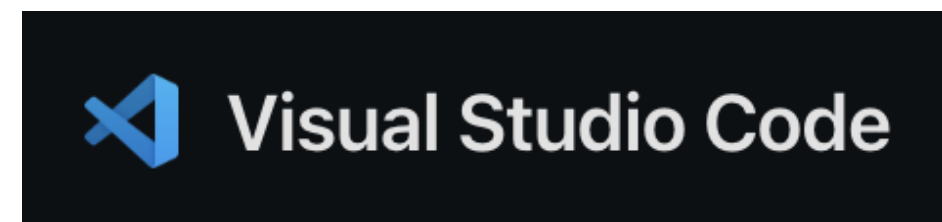
- GitHub

- Conda (Anaconda-Navigator)

- python

- Interface (Visual Studio Code  
or Jupyter Notebook)

- Text editor (Sublime Text  
or BBEdit  
or other)



# Conda

Conda is an open-source package and environment manager widely used by scientists and machine learning practitioners.

Conda has several commands that will allow you to manage your python packages:

- `conda list` will list the packages you have installed in your current environment.
- `conda install PACKAGE` will install a package (just replace `PACKAGE` with the name of the package you want to install)
- `conda update PACKAGE` will update a package
- `conda remove PACKAGE` will remove a package

For both the install, update and remove commands, you can supply a list of more than one package.

## Working with environments

It's a good idea to create a specific python environment for each of your projects. You can then save the information about your environment in an environment.yml file (more about how to use those files later in the course).

To create a new environment, you will run the following command:

```
conda create --name MYENV
```

this will create a new environment called 'MYENV'.

You might also want to specify the version of python that you would like to use in your environment. For example you need to use Python 3.9, in which case you would issue the following command:

```
conda create --name MYENV python=3.9
```



# Working with environments

To use your new environment, you need to activate it:

```
conda activate MYENV
```

Once you have activated your environment, you will see it in parentheses at the beginning of the command line prompt:

A screenshot of a macOS terminal window. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left and a title text 'OEAS805\_envdatasci — -bash — 80x24' on the right. The terminal content shows two lines of text. The first line is '(base) MacBook-Pro:OEAS805\_envdatasci jsturner\$ conda activate MYENV'. The second line is '(MYENV) MacBook-Pro:OEAS805\_envdatasci jsturner\$' followed by a grey rectangular cursor block.

```
(base) MacBook-Pro:OEAS805_envdatasci jsturner$ conda activate MYENV  
(MYENV) MacBook-Pro:OEAS805_envdatasci jsturner$
```

# Working with environments

To share your environment, 2 options:

1. Save it to a .txt file. Do this while you are working within the environment!

```
conda list --explicit > myenv.txt
```

2. Share your environment in a .yaml file, which is a text file in data serialization language, YAML, commonly used for writing configuration files. Again, you will do this while you are working within the environment you wish to export.

```
conda env export > MYENV.yaml
```

Once you have finished your session working in your environment, then you can deactivate it:

```
conda deactivate
```

# Exploratory Data Analysis

**We will be using:**

notebooks

EDA\_example.ipynb

data

city\_temperature.csv